Kunal Mishra, Ph.D.

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Bio

Computational biologist with expertise in systems genetics and multi-omics integration, spanning bulk, single-cell, and spatial transcriptomics. I lead kidney-focused research from computational discovery to experimental validation, with a focus on identifying causal genes and pathways in chronic kidney disease.

Skills

- **Computational and Programming:** R, Python, Bash, Snakemake, Git; reproducible workflow development, statistical modelling, data visualisation, high-performance computing environments
- **Omics and Systems Biology:** Bulk, single-cell, and spatial transcriptomics; multi-omics integration; systems genetics, QTL mapping, genetic association analysis, metabolomics
- **Experimental:** Cell culture, qPCR, IF/IHC, RNA-seq library preparation (bulk and single-cell); mouse and rat models

Education

- **Doctor of Philosophy (Ph.D.) in Computational Biology** — *Aug 2021 - Nov 2025*
  Centre for Computational Biology, Duke-NUS Medical School — Singapore
  - Analyzed and integrated multi-omics datasets to study the role of macrophages in chronic kidney disease (the subject of my thesis
  - Performed all computational analyses and end-to-end wet lab validation of computational findings
- **Bachelor of Science (BSc. Hons.) in Biological Sciences and Medicinal Chemistry** — *Aug 2017 - Jul 2021*
  Nanyang Technological University — Singapore
  - Graduated with Distinction, Cumulative GPA: 4.42 / 5.00
  - Relevant Coursework: Bioinformatics Methods I & II, Metabolomics and Lipidomics, Advanced Immunology, Molecular & Cell Biology I & II, Organic Chemistry I-III, Drug Discovery and Development

Experience

- **Computational Research Fellow** — *Dec 2026 - Present*
  Behmoaras Lab, Centre for Computational Biology, Duke-NUS Medical School — Singapore
  - Lead kidney-focused research projects integrating systems genetics, multi-omics analysis, and experimental validation to identify causal genes and pathways in chronic kidney disease
  - Develop and apply reproducible computational pipelines for bulk, single-cell, and spatial transcriptomic analyses
  - Communicate findings through internal and external presentations and through first-author and co-author manuscripts for peer-reviewed publication
- **Bioinformatics Research Intern** — *May 2020 - Aug 2021*
  *Prabhakhar Lab, Genome Institute of Singapore, A*STAR* — Singapore
  - Developed and improved the performance of a novel single-cell feature selection algorithm (DUBStepR, Nature Communications, 2021)
  - Analysed single-nuclei transcriptomes from human fetal brains (Simons Foundation Autism Research Initiative-funded project)
  - Analyzed pilot Rheumatoid Arthritis Single-Cell RNA Sequencing data

Research Work

‡ DENOTES EQUAL CONTRIBUTION

Publications

[1] Wong BH, **Mishra K**, Chin CF, Galam DLA, Tan BC, Behmoaras J, Chua AWC, Silver DL. **Mfsd2a is important for maintaining epidermal homeostasis** *(2026)*. *PNAS*, 123(8). DOI: 10.1073/pnas.2531159123

[2] Chen T‡, Quinn Z‡, **Mishra K**‡, O'Connor EC‡, Silver SE, Zhang Y, Valencia MJ, Mei Y, Behmoaras J, Ferreira LMR, Chatterjee P. **moPPIt: De novo generation of motif-specific and functionally active peptide binders via discrete flow matching** *(2026)*. *bioRxiv*. doi:10.1101/2024.07.31.606098

[3] Yen YH, Chew G, Chong CM, Nithianandam V, Kang PJ, Wee H, Yuan F, **Mishra K**, Behmoaras J, Cheung C, Zeng L, Su H, Ng ASL, Feany MB, Zhang SC. **FLT1-ICD loss orchestrates endothelial senescence and vascular dysfunction in Alzheimer's disease** *(2025)*. (In Review)

[4] Ouyang JF[‡], **Mishra K**[‡], Xie Y, Park H, Huang KY, Petretto E, Behmoaras J. **Systems-level identification of a matrisome-associated macrophage polarisation state in multi-organ fibrosis** *(2023)*. *eLife*, 12:e85530. DOI: 10.7554/eLife.85530

[5] Ranjan B, Sun W, Park J, **Mishra K**, Schmidt F, Xie R, Alipour F, Singhal V, Joanito I, Honardoost MA, Yong JMY, Koh ET, Leong KP, Rayan NA, Lim MGL, Prabhakar S. **DUBStepR is a scalable correlation-based feature selection method for accurately clustering single-cell data** *(2021)*. *Nature Communications*, 12:5849. DOI: 10.1038/s41467-021-26085-2

Conference Presentations

On profibrotic SPP1$^+$macrophages in disease

- **6th CSI-JSI-KAI-SgSI Joint Symposium** *2026*
  Poster. Seoul, South Korea
- **Human Cell Atlas (General Meeting)** *2025*
  Poster. Singapore
- **2nd Spatial Biology Congress: Asia** *2024*
  Poster. Singapore

Systems Genetics for Diabetic Nephropathy

- **American Society of Nephrology (Kidney Week)** *2025*
  Poster. Houston, USA
- **Advances in Cardio-Renal-Metabolic Diseases (Duke-NUS)** *2024*
  Oral Talk. Singapore

Re-purposing anti-cancer compounds as anti-malarials:

- **International Conference of Undergraduate Research** *2019*
  Oral Talk. Virtual
- **Australasian Conference for Undergraduate Research** *2019*
  Oral Talk. Newcastle, Australia

Honors and Awards

- **Centre for Technology and Development Grant (SGD 50,000)** *2026*
  Duke-NUS Medical School
  ◦ Title: *Design and optimization of lead peptide candidates for fibrotic diseases*
- **Best Lightning Talk – Advances in Cardio-renal-metabolic Diseases** *2024*
  Duke-NUS Medical School
  ◦ Title: *Systems Genetics in Diabetic Nephropathy for Causal Gene Identification*
- **Best Poster Award – Ph.D. Student Symposium** *2023*
  Duke-NUS Medical School
  ◦ Title: *Understanding the spatiotemporal activity and role of SPP1+ profibrotic macrophages in kidney disease*
- **Khoo Pre-Doctoral Fellowship** *2021*
  Duke-NUS Medical School
  ◦ Scholarship funding 4 years of stipend support and tuition
- **NISTH-Micron Responsible AI Grant (SGD 5,000)** *2019*
  Nanyang Technological University
  ◦ Title: *Development of an interactive device for the elderly for prediction of dementia*
- **Best Oral Presentation – Australasian Conference of Undergraduate Research** *2019*
  The University of Newcastle
  ◦ Title: *Identification of novel inhibitors for the malaria parasite Plasmodium falciparum*
  ◦ Received a travel grant (SGD 1,000) from NTU to present research overseas
- **Presidential Research Scholar** *2019*
  Nanyang Technological University
  ◦ Awarded for undergraduate research accomplishments with merit
  ◦ Title: *Identification of novel inhibitors for the malaria parasite Plasmodium falciparum*
- **Mr. and Mrs. Kwok Chin Yan Award for Student Initiative (SGD 1,000)** *2019*
  Nanyang Technological University
  ◦ Awarded for the *Code For Good* initiative by the NTU Open Source Society
  ◦ Led as President (2019–2020)

Media

- **SGH to lead $5.8m research project to combat deadly autoimmune disease** *2024*
  The Straits Times; Singapore General Hospital (SGH) [🌐] [🌐]
- **Scanning scars, Duke-NUS scientists find a ‘scarcode’ common across organs** *2023*
  Duke-NUS NewsHub [🌐]

Additional Information

Languages: English (Proficient), Hindi (Proficient), Indonesian (Conversational), French (Elementary)
Interests: Music (Guitar, Classical Flute), Photography